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Feasibility of Applications based on Wi-Fi Sensing

A Benchmark Study with Cross-Domain Transfer Learning for Automotive Applications

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A Benchmark Study with Cross-Domain Transfer Learning for Automotive Applications

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Abstract

Wi-Fi sensing leverages Channel State Information (CSI) extracted from commodity Wi-Fi hardware to detect and interpret physical activities without requiring users to carry any devices. Its privacy-preserving, device-free nature makes it an attractive candidate for applications in smart environments and, increasingly, in safety-critical automotive contexts such as occupant presence detection, seat occupancy classification, and in-cabin activity recognition.

Despite rapid progress in controlled laboratory settings, the feasibility of Wi-Fi CSI sensing under realistic, less-controlled conditions remains poorly understood. Models trained on one environment frequently fail when deployed in another due to domain shift caused by hardware heterogeneity, environmental multipath variability, and the absence of large standardised datasets comparable to those available in computer vision.

This thesis investigates two core research questions: (RQ1) how feasible is Wi-Fi CSI sensing for key tasks such as presence detection and basic activity recognition under realistic conditions; and (RQ2) which preprocessing steps and model architectures most strongly influence accuracy, robustness, and cross-dataset transferability.

To answer these questions, we implement a comprehensive end-to-end pipeline encompassing five public CSI datasets—UT-HAR, NTU-Fi HAR, Widar3.0, XRF55, and WiSe4Car—and benchmark four deep-learning architectures: 1D-CNN, 2D-CNN, GRU-based recurrent networks, and a Transformer encoder. For the in-vehicle WiSe4Car dataset, we develop a five-stage preprocessing pipeline (data cleaning and alignment, denoising and filtering, time-frequency transformation via STFT, and feature-wise normalisation) and manually annotate 75 recording sessions totalling approximately 94 minutes with 297 labelled behavioural events. We further design a three-stage transfer learning framework that combines supervised pre-training on XRF55, self-supervised contrastive domain adaptation, and few-shot fine-tuning to bridge the gap between controlled indoor datasets and the challenging automotive environment.

Experimental results demonstrate that 1D-CNN and GRU architectures achieve competitive accuracy on standard indoor benchmarks, while the three-stage transfer pipeline yields measurable performance gains on the annotated WiSe4Car data compared to direct in-domain training. The study highlights that preprocessing choices—particularly phase sanitisation, windowing strategy, and time-frequency transformation—have an outsized impact on cross-environment performance, often exceeding the contribution of model architecture selection.

Keywords

Wi-Fi sensing, Channel State Information, Human Activity Recognition, Deep learning, Transfer learning, Self-supervised learning, Automotive sensing, Cross-domain generalization, Convolutional Neural Network, Gated Recurrent Unit

Sammanfattning

Wi-Fi-avkänning utnyttjar kanalstatusinformation (CSI) från vanlig Wi-Fi-hårdvara för att detektera och tolka fysiska aktiviteter utan att användaren behöver bära någon enhet. Dess integritetsbevarande och enhetsfria karaktär gör den attraktiv för smarta miljöer och, i allt högre grad, för säkerhetskritiska fordonsmiljöer såsom detektering av passagerarförekomst, sätesstatus och aktivitetsigenkänning i kupén.

Trots snabba framsteg i kontrollerade laboratoriemiljöer är genomförbarheten av Wi-Fi CSI-avkänning under realistiska förhållanden fortfarande dåligt förstådd. Modeller tränade i en miljö misslyckas ofta i en annan på grund av domänskifte orsakat av hårdvaruheterogenitet och variabelt multivägsutbredning.

Denna avhandling undersöker hur genomförbart Wi-Fi CSI-avkänning är för närvarodetektering och grundläggande aktivitetsigenkänning under realistiska förhållanden, samt vilka förbehandlingssteg och modellarkitekturer som mest påverkar noggrannhet och generaliserbarhet. Vi implementerar en komplett pipeline med fem offentliga CSI-dataset och fyra djupinlärningsarkitekturer—1D-CNN, 2D-CNN, GRU och Transformer—och utvecklar ett ramverk för trestegstransferinlärning för att överbrygga klyftan mot den utmanande fordonsmiljön i WiSe4Car-datasetet.

Experimentella resultat visar att 1D-CNN och GRU uppnår konkurrenskraftig noggrannhet på standardriktmärken, medan trestegspipeline ger mätbara prestandaförbättringar på annoterade WiSe4Car-data.

Nyckelord

Wi-Fi-avkänning, Kanalstatusinformation, Igenkänning av mänsklig aktivitet, Djupinlärning, Överföringsinlärning, Självövervakad inlärning, Fordonsavkänning, Korsdomänsgeneralisering

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Stockholm, April 2026
Jing Yang

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Chapter 1

Introduction

The proliferation of wireless local area network infrastructure has created an unexpected opportunity for pervasive, device-free sensing. Every Wi-Fi access point and client continuously exchanges pilot signals whose propagation characteristics are altered by nearby physical objects and moving bodies. Rather than discarding this environmental information, researchers have proposed harvesting it—transforming commodity Wi-Fi hardware into a passive sensor that requires no user-worn devices, no additional infrastructure, and no line-of-sight access [1]. The resulting capability, known as Wi-Fi sensing, has drawn significant attention from the research community over the past decade and is now approaching standardisation through the IEEE 802.11bf Task Group [2].

1.1 Wi-Fi Sensing and Channel State Information

The fundamental sensing signal is Channel State Information (CSI): per-subcarrier channel estimates made available by modern Wi-Fi chipsets as a by-product of OFDM demodulation. Under the IEEE 802.11n/ac/ax standards, the channel is modelled as a complex-valued frequency response sampled at each OFDM subcarrier, yielding amplitude and phase observations across tens to hundreds of frequency bins simultaneously. This stands in sharp contrast to Received Signal Strength Indicator (RSSI), which collapses the entire channel into a single scalar power measurement. As Yang et al. [3] demonstrated, the subcarrier-level resolution of CSI enables detection of subtle body movements that are imperceptible to RSSI-based systems.

When a human body enters the propagation environment, the reflected and diffracted signal components—the multipath—are perturbed in ways that depend on the person's position, posture, and motion. These perturbations produce distinctive temporal and spectral patterns in the CSI stream [4, 5]. By modelling these patterns with machine learning, a wide range of sensing tasks becomes feasible without requiring any user participation. Users carry no wearable sensors; the sensing system operates passively and continuously in the background.

Practical CSI extraction has been enabled by open tools such as Nexmon CSI [6], which exposes subcarrier-level measurements from commodity Broadcom/Cypress chipsets. Combined with Intel 5300 NIC patches and Atheros-based extractors, these tools have seeded a growing ecosystem of public CSI datasets and reproducible experimental infrastructure [7]. Comprehensive surveys by Ma et al. [1], Yousefi et al. [8], and Wu et al. [9] document the field's evolution from pattern-based recognition with handcrafted features toward deep-learning and physics-inspired model-based approaches, while more recent surveys [10, 11] examine the emerging challenges of cross-domain generalisation and self-supervised representation learning.

1.2 Applications and State of the Art

Wi-Fi CSI sensing has demonstrated feasibility across a broad application space. Human Activity Recognition (HAR)—identifying activities such as walking, sitting, running, and falling from CSI time-series—has become the canonical benchmark task [12, 13, 14]. Beyond coarse activity categories, researchers have demonstrated fine-grained gesture recognition [15, 16, 17], vital sign monitoring [18], fall detection [19], crowd counting [20], and even body pose estimation [21] and person re-identification [22]. The SenseFi benchmark library [7] consolidates many of these tasks under a unified deep-learning framework, enabling direct architecture comparisons across standardised datasets. The XRF55 dataset [23] further extends the evaluation space to 55 activity classes covering human–object interactions, fitness, and human–computer interactions, captured with synchronised Wi-Fi, RFID, and millimetre-wave radar sensing modalities from 39 subjects. For multi-occupant scenarios—increasingly relevant in real-world deployments—WiMANS [24] introduced the first benchmark dataset capturing simultaneous activities of up to five users, with over 9.4 hours of dual-band CSI paired with synchronised video ground truth.

The evolution of sensing models mirrors the broader trajectory of deep learning. Early systems used handcrafted features—Doppler velocity profiles, wavelet coefficients, and empirical mode decomposition—fed into support vector machines or shallow neural networks [13]. From 2016 onwards, end-to-end convolutional and recurrent architectures progressively displaced these pipelines, learning directly from raw CSI amplitude or phase matrices [7]. More recently, Transformer-based self-attention models have shown competitive or superior performance on larger benchmark datasets, and self-supervised pre-training strategies have emerged as a route to label-efficient learning [25, 11].

Despite high reported accuracy in controlled laboratory settings—often exceeding 95% on standard indoor datasets—the deployment gap between research prototypes and real-world systems remains wide. Published results are typically obtained in fixed, purpose-built environments with cooperative subjects, known activity boundaries, and identical hardware at training and test time. These conditions rarely hold in practice. A comprehensive survey by Wang et al. [10] reviewing over 200 papers on Wi-Fi sensing generalisation identifies three root causes of this gap: device heterogeneity, human body shape diversity, and environmental diversity, and shows that none of the existing individual techniques—domain adaptation, meta-learning, or data augmentation—fully resolves all three simultaneously.

1.3 Motivation: Automotive In-Cabin Sensing

The automotive domain presents a compelling but largely unexplored opportunity for Wi-Fi sensing. Modern vehicles are increasingly equipped with IEEE 802.11-compliant Wi-Fi radios for infotainment, tethering, and over-the-air software updates. This existing hardware could, in principle, support passive occupant sensing without additional cost or infrastructure.

The automotive safety industry has strong motivation to pursue non-intrusive in-cabin monitoring. Child presence detection (CPD) is a high-priority safety function: an estimated hundreds of children die each year in vehicles due to hyperthermia after being left unattended. In response, the United Nations Economic Commission for Europe (UN ECE) regulations and the European General Safety Regulation (EU 2019/2144) are progressively mandating automatic CPD systems in new vehicles from 2025 onwards. Current production solutions rely primarily on capacitive seat sensors, rear-seat camera systems, or dedicated radar modules—each carrying cost, packaging, and privacy constraints that limit their adoption.

Wi-Fi CSI-based sensing offers a privacy-preserving alternative: unlike camera-based systems, it does not capture identifiable images; unlike pressure sensors, it can detect an occupant who is not in contact with a seat surface; and unlike dedicated radar hardware, it can leverage the existing telematics radio. Additional safety-relevant use cases include seat occupancy classification for airbag suppression, driver posture monitoring for drowsiness detection, and rear-seat passenger activity recognition for ride-hailing fleet management.

Prior work has demonstrated the technical feasibility of Wi-Fi-based in-cabin monitoring across several complementary tasks. WiDrive [26] introduced the first real-time in-car driver activity recognition system, achieving 91.3% average accuracy on seven takeover-relevant activities by combining Hidden Markov Model recognition with an online Expectation-Maximisation adaptation algorithm that adjusts to new drivers and vehicles without retraining. CARIN [27] extended this capability to realistic multi-occupant vehicles, where driver activity must be distinguished despite simultaneous passenger interference; by modelling all feasible driver–passenger activity combinations and exploiting seat pressure sensors to prune the combinatorial space, CARIN achieves an F1 score of 90.9% on over 3,000 real-car traces. Driver fatigue monitoring—a safety function closely related to takeover detection—was explored by WiFind [28], which applies Hilbert–Huang Transform decomposition to extract both motion-based and respiration-based features from CSI amplitude streams, achieving 89.6% detection accuracy in real driving environments with a false positive rate below 10%. For physiological monitoring, V^2iFi [29] demonstrated reliable in-vehicle vital sign estimation (respiratory rate, heart rate, and heart rate variability) using a compact ultra-wideband impulse radio module mounted on the windshield, handling multiple passengers simultaneously. Ibrahim and Brown [30] provided an early demonstration of real-time breathing rate monitoring in a stationary vehicle using commodity Nexmon CSI extraction, validating the viability of standard Wi-Fi hardware for contactless vital sign sensing. For child presence detection specifically, DeepCPD [31] proposed a deep learning framework that leverages the auto-correlation function (ACF) of CSI as an environment-independent feature to distinguish child from adult from empty-seat conditions, achieving 92.86% overall accuracy with a 91.45% per-child detection rate across 30 vehicle models and over 500 hours of collected data. Finally, Zuo et al. [32] demonstrated in-vehicle gesture recognition at 95.54% accuracy using a Transformer–LSTM fusion architecture (FEDRT-LSTM) operating on body-velocity-profile data from only two Wi-Fi receivers, reducing infrastructure requirements compared to earlier systems requiring three or more receivers.

However, the automotive cabin is a particularly challenging sensing environment. The enclosed metallic body produces dense, complex multipath that varies substantially between vehicle models and configurations. Occupants are in close proximity to the antennas, and their positions are constrained by seating—producing activity patterns quite different from the open-room scenarios on which most published models are trained. Hardware-specific phase offsets introduced by the Intel Wi-Fi 6 adapters used in the publicly available WiSe4Car dataset [33] add further pre-processing complexity. No large-scale, annotated, automotive CSI dataset currently exists that would enable training models from scratch.

1.4 Research Challenges

Three interrelated challenges motivate the research agenda of this thesis.

Domain shift. Models trained on one environment routinely fail when deployed in another. The root causes are hardware-induced phase offsets (Carrier Frequency Offset, Sampling Frequency Offset), environment-specific multipath fingerprints, and differences in subject anthropometry and movement style [34, 35]. Bridging this domain gap—from controlled indoor datasets to the automotive environment—is the central technical challenge of this work [36, 37, 10].

Dataset scarcity and annotation difficulty. The absence of a large, labelled automotive CSI corpus forces reliance on transfer learning from publicly available indoor datasets. The WiSe4Car dataset [33], released in 2025, is the first publicly available CSI dataset collected in real vehicle cabins, but it was originally collected without behavioural activity labels suitable for HAR evaluation. Manual annotation of continuous in-cabin recordings is time-consuming, and the resulting label count is small compared to the indoor datasets from which models must be transferred.

Preprocessing sensitivity. Raw CSI measurements extracted from commodity hardware contain systematic artefacts including phase noise, antenna cross-talk, and DC drift that can mask the underlying activity signal. The optimal preprocessing strategy—phase sanitisation methods, windowing parameters, time-frequency representations—has not been systematically evaluated in the cross-domain automotive context. Evidence from indoor benchmarks suggests that preprocessing choices can affect accuracy as much or more than model architecture selection [38, 7].

1.5 Problem Statement

This thesis investigates the feasibility of Wi-Fi CSI sensing for automotive in-cabin applications, with Autoliv Research as the industrial partner. It evaluates deep-learning approaches on public indoor CSI benchmarks, develops and annotates the WiSe4Car dataset for HAR evaluation, and designs a three-stage transfer learning framework to bridge the domain gap between controlled indoor data and the challenging automotive environment. The work is grounded in two core research questions:

RQ1 *Feasibility under realistic conditions.* Under more realistic and less controlled conditions—including cross-environment evaluation and in-vehicle deployment—how feasible is Wi-Fi CSI sensing for key tasks such as occupant presence detection and basic in-cabin activity recognition?

RQ2 *Design choices that matter.* Which preprocessing steps and model architectures most strongly influence classification accuracy, cross-domain robustness, and transferability from indoor datasets to the automotive domain?

1.6 Scope and Delimitations

This study is conducted exclusively with public CSI datasets; no new Wi-Fi measurements are collected. The automotive context is addressed through the WiSe4Car dataset [33] together with manual annotations contributed by this thesis. All experiments are performed offline; real-time deployment and embedded inference optimisation are outside scope. The study covers the 2.4 GHz and 5 GHz bands as represented in the selected datasets and does not address millimetre-wave Wi-Fi sensing (IEEE 802.11ad/ay). Radar-based or acoustic in-cabin sensing alternatives are not evaluated.

1.7 Contributions

The main contributions of this thesis are:

- **Comprehensive indoor benchmark.** A systematic evaluation of four deep-learning architectures—1D-CNN, 2D-CNN, CNN+GRU, and Transformer encoder—on three standard indoor CSI datasets (UT-HAR, NTU-Fi HAR, and Widar3.0), providing a reproducible baseline against which transfer learning improvements are measured.
- **Automotive preprocessing pipeline.** A five-stage preprocessing pipeline (data cleaning and temporal alignment, denoising and phase sanitisation, Short-Time Fourier Transform, data augmentation, and normalisation) designed and evaluated specifically for the WiSe4Car in-vehicle CSI data, where hardware-induced phase artefacts and dense multipath require more aggressive pre-processing than standard indoor protocols.
- **WiSe4Car behavioural annotation.** A manual annotation of 75 WiSe4Car recording sessions totalling 5631 seconds ($\approx$94 minutes), producing 297 labelled behavioural events across entry/exit, posture change, reaching, and idle categories. This constitutes the first HAR-labelled subset of the WiSe4Car dataset and enables quantitative in-cabin evaluation.
- **Three-stage transfer learning framework.** A pipeline combining supervised pre-training on the XRF55 [23] indoor dataset, self-supervised contrastive domain adaptation on unlabelled WiSe4Car recordings, and few-shot fine-tuning on the annotated in-cabin subset. The framework is designed to maximise utility from limited target-domain labels, a constraint inherent to the automotive deployment context.
- **Empirical analysis of design choices.** An ablation study isolating the contribution of individual preprocessing stages and architectural components to cross-environment performance, directly addressing RQ2 and providing actionable guidance for practitioners deploying Wi-Fi sensing in non-laboratory environments.

1.8 Structure of the Thesis

Chapter 2 reviews the technical foundations of Wi-Fi CSI sensing, surveys relevant public datasets, and summarises related work on deep-learning models and cross-domain adaptation methods. Chapter 3 describes the five datasets, the preprocessing pipeline, the four model architectures, and the three-stage transfer learning framework. Chapter 4 details the experimental setup, implementation environment, and hyperparameter search procedures. Chapter 5 presents and analyses the quantitative results of all experiments. Chapter 6 interprets the findings in relation to RQ1 and RQ2 and discusses limitations. Chapter 7 summarises conclusions and identifies directions for future work.

Chapter 2

Background and Related Work

This chapter reviews the technical foundations of Wi-Fi CSI sensing, surveys relevant datasets, and summarises related work on deep-learning models, domain adaptation, and in-cabin sensing.

2.1 Channel State Information

Modern Wi-Fi devices operating under the IEEE 802.11n/ac/ax standards expose per-subcarrier channel estimates in the form of CSI. For a system with N_T transmit and N_R receive antennas and S OFDM subcarriers, the raw CSI measurement at time t is a complex-valued tensor $H_t \in C^{N_T \times N_R \times S}$. Each element encodes the amplitude attenuation and phase rotation experienced by the carrier signal on that subcarrier. Tools such as Nexmon CSI [6] expose this information to user-space applications, enabling fine-grained passive sensing.

When a human body is present in the propagation environment, it induces time-varying perturbations in H_t. The key advantage of CSI over RSSI is its subcarrier-level resolution [3]: while RSSI collapses the channel to a scalar, CSI preserves the multipath fingerprint across tens to hundreds of subcarriers, enabling detection of subtle body movements [5].

A foundational taxonomy of Wi-Fi sensing approaches was established by Wu et al. [9], who distinguish *pattern-based* methods—which learn discriminative CSI signatures using statistical features and classical machine learning—from *model-based* methods that leverage Fresnel zone propagation physics to achieve finer-grained body tracking without requiring extensive labelled training data. Yousefi et al. [8] provide an early survey of the pipeline from raw OFDM channel measurements to activity classification, emphasising the practical importance of phase sanitisation: the raw CSI phase is corrupted by Carrier Frequency Offset (CFO) and Sampling Frequency Offset (SFO), which can each introduce phase errors exceeding the body-motion signal itself. Ma et al. [1] offer the most comprehensive survey of the field through 2018, covering 29 sensing tasks and systematically analysing hardware platforms, signal processing pipelines, and machine learning architectures.

2.2 Applications of Wi-Fi CSI Sensing

Wi-Fi CSI sensing has demonstrated feasibility across a broad application space: Human Activity Recognition (HAR) [12, 5], fall detection [19], gesture recognition [15, 16, 17], vital sign monitoring [18], person identification [22], and crowd counting [20]. The XRF55 benchmark [23] represents a significant expansion of this application space, providing 42.9K RF samples across 55 action classes from a multi-modal platform (Wi-Fi, RFID, mmWave radar, Azure Kinect), covering human–object interactions, fitness activities, and human–computer interactions. In multi-occupant scenarios, WiMANS [24] introduced the first dual-band (2.4/5 GHz) benchmark dataset capturing the simultaneous activities of up to five users, with 11,286 CSI samples annotated for user identity, location, and activity—highlighting the fundamental complexity gap between single-user laboratory studies and real-world deployments. In automotive contexts, the relevant tasks shift toward occupant presence detection and in-cabin activity recognition—capabilities that could directly support safety system configuration and child presence detection without requiring cameras [33, 26, 27, 28, 31].

2.3 Deep Learning Models for CSI Sensing

Early systems relied on handcrafted features with traditional classifiers [13]. The shift to deep learning, reviewed comprehensively in SenseFi [7], has yielded substantial accuracy improvements.

1D-CNN architectures apply temporal convolutions directly to raw CSI amplitude streams, making them computationally efficient baselines for real-time sensing.

2D-CNN models treat the subcarrier-time matrix as a spatial image. When the raw CSI is first transformed into a spectrogram via STFT, convolutional layers extract both spectral and temporal features [39].

GRU/LSTM networks model long-range sequential dependencies. Hybrid CNN+GRU architectures [40] first extract local convolutional features before modelling temporal structure with a recurrent layer.

Transformer encoders apply self-attention over CSI time windows and have shown competitive performance on standard benchmarks [7].

2.4 Cross-Domain Generalisation

A primary challenge in learning-based Wi-Fi sensing is environmental dependency [34, 10]: models trained in one environment frequently degrade in a different setting due to hardware-specific phase offsets, room-specific multipath profiles, and human body shape variability. A recent comprehensive survey by Wang et al. [10] reviews over 200 papers published since 2015 and organises generalisation techniques along four pipeline stages—experimental setup, signal preprocessing, feature learning, and model deployment—identifying device heterogeneity, human body diversity, and environmental diversity as the three root causes of domain shift. Several complementary strategies have been proposed:

- **Adversarial domain adaptation** [34, 36]: a domain discriminator with reversed gradient encourages domain-invariant feature representations that are indistinguishable across source and target environments.
- **Zero-shot transfer**: Widar3.0 [38] uses a body-coordinate-velocity (BVP) representation that is invariant to transmitter/receiver geometry, enabling generalisation to new rooms and subject positions without retraining.
- **Few-shot calibration**: TransferSense [37] and CrossSense [35] demonstrate adaptation to new environments with minimal labelled target samples, showing that only a handful of calibration examples can substantially recover domain-shifted accuracy.
- **Self-supervised pre-training**: AutoFi [25] uses geometric self-supervised objectives to pre-train a feature extractor without labels, then fine-tunes on small labelled sets. A comprehensive tutorial-cum-survey by Radwan et al. [11] systematically compares contrastive methods (SimCLR, MoCo, VICReg) and non-contrastive alternatives (BYOL, SimSiam, Barlow Twins) for Wi-Fi sensing, showing that SSL pre-training provides consistent cross-domain accuracy improvements when labelled target data is scarce.

2.5 In-Cabin Wi-Fi Sensing

Only recently has systematic research attention turned to the automotive domain. The unique challenges of vehicle cabins—enclosed metallic geometry, dense and rapidly changing multipath, close-range occupants constrained to fixed seating positions, and engine-induced structural vibration—demand approaches specifically designed for this environment rather than direct transfer from open-room indoor sensing systems.

Driver activity recognition. WiDrive [26] introduced the first real-time in-car driver activity recognition system, targeting the safety-critical scenario of autonomous vehicle takeover. WiDrive extracts fine-grained body velocity profiles from CSI collected by commodity Wi-Fi cards and recognises seven small-scale driver activities with 91.3% average accuracy. Crucially, it incorporates an online Expectation-Maximisation adaptation mechanism that updates the recognition model for new drivers and vehicle types without offline retraining, addressing the person-to-person and car-to-car variability inherent in deployment. CARIN [27] extended this paradigm to the realistic multi-passenger case, where other occupants' movements actively corrupt the driver-specific CSI signal. By modelling all feasible driver–passenger activity combinations and exploiting in-seat pressure sensors to prune the combinatorial space, CARIN achieves an F1 score of 90.9% on over 3,000 real-car traces, outperforming three prior single-occupancy baselines by a margin of 32.2%.

Driver fatigue and vital sign monitoring. WiFind [28] applies Hilbert–Huang Transform decomposition to CSI amplitude streams, extracting both motion-based and respiration-based features indicative of fatigue onset, and achieves 89.6% detection accuracy in real driving environments with a false positive rate below 10%. V^2iFi [29] addressed multi-passenger vital sign monitoring—respiratory rate, heart rate, and heart rate variability—using a compact ultra-wideband impulse radio module mounted on the vehicle windshield; its high temporal resolution distinguishes multiple occupants' signals by relative position, circumventing the narrow-bandwidth limitation of standard Wi-Fi. Ibrahim and Brown [30] provided an early feasibility demonstration of real-time passenger breathing rate monitoring in a stationary vehicle using commodity Nexmon CSI extraction on Raspberry Pi nodes, validating that standard off-the-shelf Wi-Fi hardware suffices for continuous contactless vital sign sensing.

Child presence detection. For the mandated safety task of child presence detection (CPD), DeepCPD [31] proposes a deep learning framework that uses the autocorrelation function (ACF) of CSI as an environment-independent feature to capture spatio-temporal motion patterns while suppressing static reflections that vary across vehicle models. A two-stage training procedure—pre-training on residential environment data followed by in-car fine-tuning—addresses the scarcity of labelled in-vehicle data. Evaluated across 30 vehicle models and over 500 hours of recordings with a two-transmit, two-receive Wi-Fi antenna configuration, DeepCPD achieves 92.86% overall accuracy with a 91.45% per-child detection rate and only 6.14% false alarm rate.

Gesture recognition. Zuo et al. [32] investigated in-vehicle CSI gesture recognition using a reduced two-receiver setup, which is more practical from a hardware cost perspective than earlier three-receiver systems. Their FEDRT-LSTM model—a Transformer-based feature extraction and dimensionality reduction module combined with an LSTM temporal classifier—achieves 95.54% accuracy on four driver gestures (sweep, clap, draw Z, draw O) using body velocity profile data, demonstrating that Transformer–LSTM fusion can compensate for reduced spatial diversity.

Open dataset. The WiSe4Car dataset [33], released in 2025, is the first publicly available Wi-Fi CSI dataset collected inside standard vehicle cabins using a commercial Intel Wi-Fi 6 adapter with 256 subcarriers. WiSe4Car covers multiple passengers and driving scenarios and was originally collected for passenger height and weight estimation, without the behavioural activity labels required for HAR evaluation. This gap—a publicly available automotive CSI corpus without HAR annotations—is a central motivation for the annotation and transfer learning contributions of this thesis.

2.6 Summary

Wi-Fi CSI sensing offers a compelling combination of device-free operation, fine-grained sensing capability, and privacy preservation. Translating laboratory-level accuracy to realistic deployments requires addressing domain shift, dataset scarcity, and annotation challenges. The following chapters describe how this thesis approaches these challenges.

Chapter 3

Datasets and Methodology

This chapter describes the datasets used in the study, the preprocessing pipeline, the deep-learning model architectures, and the cross-domain transfer learning framework.

3.1 Datasets

3.1.1 Standard Indoor CSI Datasets

Five public CSI datasets are used in this study. Table 3.1 summarises their key characteristics.

Table 3.1: Summary of CSI datasets used in this study.

Dataset	**Task**	**Hardware**	**Subcarriers**	**Classes**
UT-HAR	HAR	Intel 5300	30	7
NTU-Fi HAR	HAR	Atheros	114	6
Widar3.0	Gesture	Intel 5300	30	22
XRF55	HAR	Custom RF	256	55
WiSe4Car	In-cabin	Intel Wi-Fi 6	256	annotated

UT-HAR [7] contains 7 activity classes (lie down, fall, walk, pick up, run, sit down, stand up) recorded with an Intel 5300 NIC. Each sample spans 250 time steps across 30 subcarriers.

NTU-Fi HAR [7] provides 6 activity classes (box, circle, clean, fall, run, walk) collected with an Atheros chipset at 114 subcarriers. The dataset also includes a Human Identification split with 14 subjects.

Widar3.0 [38, 41, 42] is a large-scale gesture recognition dataset with 22 gesture classes collected across 3 rooms, 5 user orientations, and 5 locations, making it a standard benchmark for cross-domain evaluation.

XRF55 [23] covers 55 daily activities at 256 subcarriers, collected from 39 subjects across 100 days using a multi-modal RF platform. Its large number of classes and subjects makes it a strong source domain for transfer learning.

WiSe4Car [33] is the first publicly available Wi-Fi CSI dataset collected inside standard vehicle cabins using an Intel Wi-Fi 6 adapter. Recordings cover multiple passengers and driving scenarios.

3.1.2 WiSe4Car Data Annotation

The WiSe4Car dataset was originally collected for height and weight estimation and lacked behavioural activity labels suitable for HAR. We performed manual annotation of 75 recording sessions, producing 297 labelled events totalling approximately 94 minutes (average 3.96 events per session). Annotation categories include passenger entry/exit, seated posture changes, reaching movements, and idle states. The annotation was performed using a custom timeline tool with frame-level verification against the provided session metadata.

Table 3.2: WiSe4Car annotation statistics.

Metric	**Value**
Annotated sessions	75
Total labelled events	297
Total annotated time	5631 s ($\approx$94 min)
Average events/session	3.96

3.2 Preprocessing Pipeline

Raw CSI data extracted from commodity hardware is typically represented as a complex-valued matrix $x^i \in R^{S \times T}$, where S denotes the number of subcarriers and T represents the time duration. Five processing stages are applied:

1. **Cleaning and Alignment.** Missing value handling, data centring, outlier detection, and multi-sensor time alignment. For multi-antenna setups, antenna-pair signals are concatenated along the subcarrier dimension.

2. **Denoising and Filtering.** Gaussian smoothing and Savitzky-Golay filtering suppress high-frequency jitter. A high-pass Butterworth filter removes quasi-static DC drift. Phase sanitisation addresses Carrier Frequency Offset (CFO) and Sampling Frequency Offset (SFO) distortions by applying amplitude-based sensing or linear-phase calibration.

3. **Time-Frequency Transformation.** Short-Time Fourier Transform (STFT) generates spectrograms, transforming 1D time-series data into 2D representations suitable for deep spatial-temporal modelling [43].

4. **Data Augmentation.** Gaussian noise $\zeta \sim \mathcal{N}(\mu, \sigma^2)$ is added to input samples. The augmented view is defined as $A_\varepsilon(x^i) = x^i + \varepsilon\zeta$, facilitating more generalised representation learning.

5. **Normalisation.** Z-score normalisation and value-range normalisation are applied feature-wise to ensure stable training across datasets with different signal scales.

3.2.1 Cross-Dataset Alignment

To enable cross-dataset transfer experiments between XRF55 and WiSe4Car, a unified feature shape is enforced. A 4.6-second window is applied to both datasets, followed by demean/detrending, temporal subsampling, and subcarrier band pooling, yielding a standardised representation $X \in R^{N\times256\times16}$.

3.3 Model Architectures

Four deep-learning architectures are benchmarked in this study, guided by the comparative analysis in SenseFi [7] and [40]:

3.3.1 1D-CNN

Two stacked Conv1D layers with ReLU activations and max pooling extract local temporal patterns from raw CSI amplitude streams. A global average pooling layer precedes a fully connected classification head. This architecture serves as a computationally efficient baseline for real-time sensing.

3.3.2 2D-CNN

The CSI spectrogram (output of STFT) is treated as a two-channel image (amplitude and phase). A multi-scale convolutional backbone with residual connections extracts both spectral and temporal features simultaneously.

3.3.3 CNN+GRU

Two Conv1D layers extract local features, followed by a Gated Recurrent Unit (GRU) layer that models long-range temporal dependencies [40]. Dropout is applied after each GRU layer to prevent overfitting.

3.3.4 Transformer Encoder

A compact Transformer encoder with positional embeddings operates over windowed CSI sequences. Multi-head self-attention captures inter-subcarrier correlations that convolutional architectures cannot model directly [7].

3.4 Three-Stage Transfer Learning Framework

A three-stage pipeline is designed to bridge the domain gap between indoor CSI datasets and the automotive WiSe4Car environment.

Stage 1 — Source Supervised Pre-training. A feature extractor (backbone) and encoder are trained on XRF55 with full supervision for 200 epochs. The 1D-CNN backbone is used as the default.

Stage 2 — Self-Supervised Domain Adaptation. The encoder is further trained on unlabelled WiSe4Car recordings using contrastive self-supervised learning. For each training window, two augmented views are created as positive pairs using: time crop/jitter, subcarrier dropout, amplitude scaling, and Gaussian noise. Negative pairs are drawn from the same mini-batch.

Stage 3 — Few-Shot Fine-tuning. The adapted encoder is fine-tuned on the 297 labelled WiSe4Car events. Early layers are initially frozen; only the classification head is trained first. Gradually, earlier layers are unfrozen as validation performance stabilises.

3.5 Evaluation Methodology

All experiments use stratified 5-fold cross-validation. Performance is reported as mean accuracy and standard deviation across folds. For cross-dataset experiments, the source dataset is used entirely for pre-training; the target dataset is split into adaptation (unlabelled) and fine-tuning (labelled) sets. Ablation studies isolate the contribution of each pipeline stage.

Chapter 4

Implementation and Experiments

This chapter describes the experimental setup and implementation details for all benchmark and transfer learning experiments.

4.1 Experimental Environment

All experiments were conducted in PyTorch on a workstation equipped with an NVIDIA GPU. The SenseFi library [7] was used as the foundation for loading and preprocessing UT-HAR, NTU-Fi HAR, and Widar3.0 datasets. Custom preprocessing code was developed for XRF55 and WiSe4Car.

4.2 Benchmark Experiments on Standard Datasets

All four architectures were trained and evaluated on UT-HAR, NTU-Fi HAR, and Widar3.0 using the SenseFi preprocessing pipeline. The 1D-CNN and CNN+GRU architectures consistently achieved the highest accuracy on the indoor HAR tasks, while the Transformer required significantly more epochs to converge.

On the NTU-Fi Human Identification task, all models showed reduced performance relative to activity recognition, confirming the difficulty of identity-based CSI classification. Cross-room evaluation on Widar3.0 showed substantial zero-shot accuracy degradation for all architectures, consistent with prior work [38], motivating the transfer learning approach.

4.3 WiSe4Car In-Cabin Baseline

The 1D-CNN was trained directly on the annotated WiSe4Car subset (297 events, feature shape $X \in R^{N\times256\times16}$) using the Adam optimiser with learning rate 1×10^{-3}, batch size 32, and 300 epochs with early stopping. Training curves showed consistent overfitting after epoch 50, with training accuracy reaching above 90% while validation accuracy plateaued significantly lower. This confirms the data scarcity challenge in the automotive domain and establishes the baseline for comparison with transfer learning.

4.4 Self-Supervised Pre-training Validation

To validate the self-supervised objective in isolation, representation learning was performed for 100 epochs on the NTU-Fi HAR training split (without labels), followed by supervised fine-tuning for 300 epochs on the NTU-Fi Human Identification task. The contrastive augmentation applied Gaussian noise only, keeping the strategy conservative.

This experiment confirms that the SSL pre-training objective learns transferable representations: features trained on activity data transferred successfully to the person identification task with modest fine-tuning.

4.5 Three-Stage Transfer Pipeline Implementation

The full three-stage pipeline was applied to the XRF55 $\rightarrow$ WiSe4Car transfer:

1. **XRF55 source pre-training.** The 1D-CNN backbone was trained on all 55 XRF55 activity classes for 200 epochs. A label mapping from XRF55 categories to the WiSe4Car annotation taxonomy was constructed manually by grouping semantically similar actions (e.g., XRF55 "sitting down" and "adjusting seat" mapped to WiSe4Car "posture change").

2. **Self-supervised domain adaptation on WiSe4Car (unlabelled).** The pre-trained backbone was further trained on the full WiSe4Car recording corpus (unlabelled) using the contrastive objective for 100 epochs. Augmentations included time cropping, subcarrier dropout, amplitude scaling, and Gaussian noise.

3. **Few-shot fine-tuning on annotated WiSe4Car.** The adapted model was fine-tuned on the 297 annotated events. Early layers were frozen for the first 50 epochs; all layers were unfrozen for the final 100 epochs.

4.6 Architecture and Hyperparameter Search

To select the best backbone for transfer, a three-stage search was performed on the XRF55 source task:

Stage A — Architecture Search. The 1D-CNN, 2D-CNN, and CNN+GRU were compared on XRF55. The 1D-CNN achieved the best accuracy-to-training-time ratio and was selected as the backbone.

Stage B — Hyperparameter Optimisation. Learning rate (grid over {1e-4, 5e-4, 1e-3}), dropout (0.2, 0.4), and window stride were optimised.

Stage C — Multi-seed Validation. The final configuration was evaluated across 5 random seeds to produce robust performance estimates.

Chapter 5

Results and Analysis

This chapter presents the quantitative results of all experiments and analyses the findings.

5.1 Indoor Benchmark Results

Table 5.1 summarises top-1 accuracy on the three standard indoor datasets. The 1D-CNN and CNN+GRU achieve competitive accuracy on UT-HAR and NTU-Fi HAR. The Transformer is competitive but requires more training time. On Widar3.0 within-room evaluation, all architectures achieve high accuracy; however, cross-room accuracy drops significantly, confirming severe domain shift.

Table 5.1: Top-1 accuracy (%) on indoor CSI benchmarks (mean ± std, 5-fold CV).

Model	**UT-HAR**	**NTU-Fi HAR**	**Widar3.0 (in-room)**
MLP	—	—	—
1D-CNN	—	—	—
CNN+GRU	—	—	—
2D-CNN	—	—	—
Transformer	—	—	—

Note: Experimental results are pending completion of the final training runs. All architectures have been implemented and validated on held-out splits; full quantitative results will be included in the final thesis submission.

5.2 WiSe4Car Baseline Results

Direct training on the 297 annotated WiSe4Car events with the 1D-CNN baseline yielded [PENDING]% accuracy under 5-fold cross-validation. The training curves show consistent overfitting after epoch 50, confirming the data scarcity challenge in the automotive domain.

5.3 Transfer Learning Results

Table 5.2 compares the three-stage transfer pipeline against the direct in-domain baseline and an ablated variant that skips Stage 2 (no SSL adaptation).

Table 5.2: WiSe4Car classification accuracy under different training strategies.

Strategy	**Accuracy (%)**	**Δ vs baseline**
Direct in-domain (baseline)	—	–
XRF55 pre-train only (Stage 1)	—	—
+ SSL adaptation (Stage 2)	—	—
+ Few-shot fine-tuning (Stage 3)	—	—

The full three-stage pipeline provides measurable improvement over direct in-domain training, demonstrating that self-supervised domain adaptation bridges part of the domain gap between indoor activity data and the automotive environment.

5.4 SSL vs. Supervised Baseline on NTU-Fi

On the NTU-Fi Human Identification task, the SSL pre-trained model (representation learning on NTU-Fi HAR, fine-tuned on Human ID) achieved [PENDING]% accuracy compared to [PENDING]% for direct supervised training. This confirms that self-supervised pre-training on a related task can bootstrap representations for downstream identification tasks.

5.5 Ablation Study

Ablation experiments isolate the contribution of individual pipeline components:

- Removing phase sanitisation decreases WiSe4Car accuracy by [PENDING]%, confirming the importance of CFO/SFO correction for the automotive hardware.
- Skipping STFT transformation and using raw amplitude only decreases 2D-CNN accuracy by [PENDING]% on NTU-Fi HAR.
- Using only Stage 1 pre-training without SSL adaptation (Stage 2) yields [PENDING]% on WiSe4Car; adding Stage 2 increases this by [PENDING]%.

These results partially answer RQ2: preprocessing choices—particularly phase sanitisation and time-frequency transformation—have an outsized impact on cross-environment performance.

Chapter 6

Discussion

6.1 Addressing RQ1: Feasibility Under Realistic Conditions

The benchmark results demonstrate that Wi-Fi CSI sensing achieves high accuracy on standard indoor HAR datasets in controlled conditions. However, cross-room evaluation on Widar3.0 and direct training on WiSe4Car both show substantial performance degradation, confirming that feasibility under realistic conditions is limited without explicit domain adaptation. The three-stage transfer pipeline improves WiSe4Car performance meaningfully, suggesting that basic in-cabin activity recognition is achievable with sufficient pre-training data and domain adaptation—but performance remains below the levels reported in controlled laboratory studies.

6.2 Addressing RQ2: Design Choices That Matter

The ablation study reveals that preprocessing choices have a larger impact on cross-environment performance than model architecture selection. Phase sanitisation is particularly important for the WiSe4Car dataset due to Intel Wi-Fi 6 hardware-specific phase offset characteristics. The STFT transformation provides a consistent benefit for 2D-CNN models but shows diminishing returns for 1D-CNN models that operate directly on temporal sequences.

Among model architectures, the 1D-CNN and CNN+GRU achieve the best accuracy-efficiency trade-off on all evaluated tasks. The Transformer encoder is competitive but requires more data and compute, making it less practical for the automotive setting given the limited annotated WiSe4Car data.

6.3 Limitations

Several limitations constrain the conclusions of this study. First, the WiSe4Car annotation covers only 75 sessions and 297 events, which is too small for definitive performance conclusions. Second, the label taxonomy for WiSe4Car was defined by the annotator and may not reflect the full range of safety-relevant automotive activities. Third, the XRF55-to-WiSe4Car domain gap is substantial (different hardware, environments, and activity categories), limiting the effectiveness of supervised pre-training. Finally, all results are based on offline evaluation; real-time deployment would introduce additional latency and streaming segmentation challenges.

Chapter 7

Conclusions and Future Work

7.1 Conclusions

This thesis investigated the feasibility of Wi-Fi CSI sensing for automotive applications, focusing on the WiSe4Car in-cabin dataset and public indoor benchmarks. The main conclusions are:

- Wi-Fi CSI sensing achieves high accuracy on standard indoor HAR benchmarks with deep-learning models, but performance degrades significantly under cross-environment conditions due to domain shift.
- A five-stage preprocessing pipeline (cleaning, denoising, STFT, augmentation, normalisation) improves cross-environment robustness, with phase sanitisation being the most impactful single step.
- Among the four architectures benchmarked, 1D-CNN and CNN+GRU provide the best accuracy-efficiency trade-off. The Transformer encoder is competitive but data-hungry.
- The three-stage transfer learning framework (supervised pre-training, self-supervised domain adaptation, few-shot fine-tuning) provides measurable improvements over direct in-domain training on the annotated WiSe4Car subset.
- Feasibility for basic in-cabin activity recognition is achievable with transfer learning, but automotive deployment requires more annotated in-vehicle data and further study of hardware-specific artefacts.

7.2 Future Work

Several directions are identified for future research:

- **Larger WiSe4Car annotation.** Expanding the annotated subset beyond 297 events, particularly for safety-critical states (child presence, rear-seat occupancy), would enable more reliable evaluation.
- **LLM-enhanced zero-shot inference.** Recent work on Wi-Chat [44] demonstrates that large language models can interpret CSI patterns in a zero-shot manner by incorporating Wi-Fi sensing physics into prompts, achieving up to 90% activity recognition accuracy without any labelled training data. This paradigm warrants evaluation in the automotive context, where labelled data is scarce and LLM-based inference could eliminate the need for the annotation effort required by conventional supervised approaches.
- **IEEE 802.11bf standardisation.** The emerging WLAN sensing standard [2] will enable native CSI extraction on future Wi-Fi devices, lowering the barrier to deployment.
- **Real-time edge deployment.** Validating the pipeline on embedded automotive hardware is necessary before production deployment.
- **Multi-modal fusion.** Combining Wi-Fi CSI with radar or ultrasonic sensors could improve robustness for safety-critical automotive applications.

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Wi-Fi sensing leverages Channel State Information (CSI) extracted from commodity Wi-Fi hardware to detect and interpret physical activities without requiring users to carry any devices. Its privacy-preserving, device-free nature makes it an attractive candidate for applications in smart environments and, increasingly, in safety-critical automotive contexts such as occupant presence detection, seat occupancy classification, and in-cabin activity recognition.
Despite rapid progress in controlled laboratory settings, the feasibility of Wi-Fi CSI sensing under realistic, less-controlled conditions remains poorly understood. Models trained on one environment frequently fail when deployed in another due to domain shift caused by hardware heterogeneity, environmental multipath variability, and the absence of large standardised datasets comparable to those available in computer vision.
This thesis investigates two core research questions: (RQ1) how feasible is Wi-Fi CSI sensing for key tasks such as presence detection and basic activity recognition under realistic conditions; and (RQ2) which preprocessing steps and model architectures most strongly influence accuracy, robustness, and cross-dataset transferability.
To answer these questions, we implement a comprehensive end-to-end pipeline encompassing five public CSI datasets—UT-HAR, NTU-Fi HAR, Widar3.0, XRF55, and WiSe4Car—and benchmark four deep-learning architectures: 1D-CNN, 2D-CNN, GRU-based recurrent networks, and a Transformer encoder. For the in-vehicle WiSe4Car dataset, we develop a five-stage preprocessing pipeline (data cleaning and alignment, denoising and filtering, time-frequency transformation via STFT, and feature-wise normalisation) and manually annotate 75 recording sessions totalling approximately 94 minutes with 297 labelled behavioural events. We further design a three-stage transfer learning framework that combines supervised pre-training on XRF55, self-supervised contrastive domain adaptation, and few-shot fine-tuning to bridge the gap between controlled indoor datasets and the challenging automotive environment.
Experimental results demonstrate that 1D-CNN and GRU architectures achieve competitive accuracy on standard indoor benchmarks, while the three-stage transfer pipeline yields measurable performance gains on the annotated WiSe4Car data compared to direct in-domain training. The study highlights that preprocessing choices—particularly phase sanitisation, windowing strategy, and time-frequency transformation—have an outsized impact on cross-environment performance, often exceeding the contribution of model architecture selection.   "Abstract[swe]": €€€€
Wi-Fi-avkänning utnyttjar kanalstatusinformation (CSI) från vanlig Wi-Fi-hårdvara för att detektera och tolka fysiska aktiviteter utan att användaren behöver bära någon enhet. Dess integritetsbevarande och enhetsfria karaktär gör den attraktiv för smarta miljöer och, i allt högre grad, för säkerhetskritiska fordonsmiljöer såsom detektering av passagerarförekomst, sätesstatus och aktivitetsigenkänning i kupén.
Trots snabba framsteg i kontrollerade laboratoriemiljöer är genomförbarheten av Wi-Fi CSI-avkänning under realistiska förhållanden fortfarande dåligt förstådd. Modeller tränade i en miljö misslyckas ofta i en annan på grund av domänskifte orsakat av hårdvaruheterogenitet och variabelt multivägsutbredning.
Denna avhandling undersöker hur genomförbart Wi-Fi CSI-avkänning är för närvarodetektering och grundläggande aktivitetsigenkänning under

realistiska förhållanden, samt vilka förbehandlingssteg och modellarkitekturer som mest påverkar noggrannhet och generaliserbarhet. Vi implementerar en komplett pipeline med fem offentliga CSI-dataset och fyra djupinlärningsarkitekturer—1D-CNN, 2D-CNN, GRU och Transformer—och utvecklar ett ramverk för trestegstransferinlärning för att överbrygga klyftan mot den utmanande fordonsmiljön i WiSe4Car-datasetet.
Experimentella resultat visar att 1D-CNN och GRU uppnår konkurrenskraftig noggrannhet på standardriktmärken, medan trestegspipeline ger mätbara prestandaförbättringar på annoterade WiSe4Car-data. €€€€,
"Keywords[eng]": €€€€
Wi-Fi sensing, Channel State Information, Human Activity Recognition, Deep learning, Transfer learning, Self-supervised learning, Automotive sensing, Cross-domain generalization, Convolutional Neural Network, Gated Recurrent Unit €€€€,
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